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AB Testguide

(/)

Is your test result significant? Does it have enough power?

Play with the controls and get a better feel for how a lower confidence level will boost the power or how an increase in test size can make a small CR-difference significant!

Pre-test calculation or post-test evaluation?

- ☐ Pre-test analysis
- ☒ Test evaluation

Test data

Visitors A

Visitors B

Conversions A

Conversions B

Settings

Hypothesis (?)

- ☒ One-sided
- ☐ Two-sided

Confidence (?)

- ☐ 90%
- ☒ 95%
- ☐ 99%

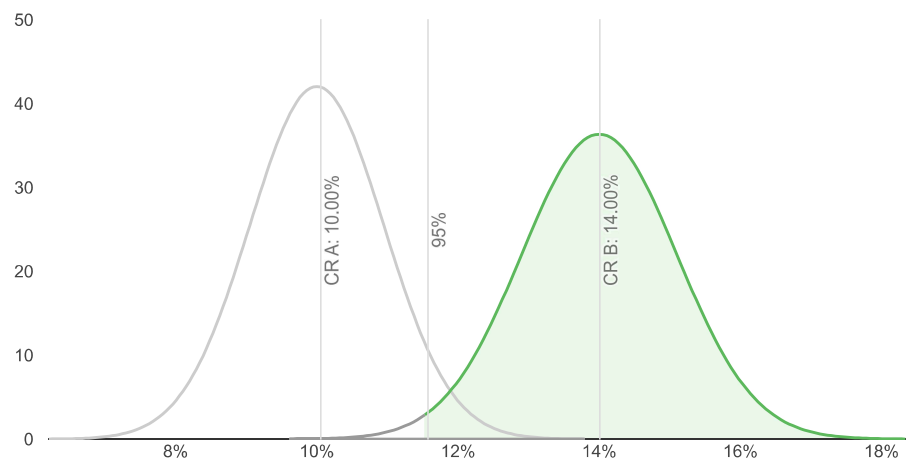
Test result

Significant test result!

Variation B's observed conversion rate (14.00%) was 40.00% higher than variation A's conversion rate (10.00%). You can be 95% confident that this result is a consequence of the changes you made and not a result of random chance.



The expected distributions of variation A and B.



Conversion Rate Control

Conversions A / Visitors A

10.00%

Conversion Rate B

Conversions B / Visitors B

14.00%

Relative uplift in Conversion Rate

$CR_B - CR_A / CR_A$

40.00%

Observed Power

98.69%

p value

0.0029

Z-score

$(CR_B - CR_A) / SE_{\text{difference}}$

2.7576

Standard error A

$(CR_A * (1 - CR_A) / \text{Visitors}_A)^{1/2}$

0.009487

Standard error B

$(CR_B * (1 - CR_B) / \text{Visitors}_B)^{1/2}$

0.010973

Std. Error of difference

$SE_{\text{difference}} = (SE_A^2 + SE_B^2)^{1/2}$

0.014505

More tools

A/B Test Size Calculator - How many visitors do you need? (<https://abtestguide.com/abtestsize/>)

Bayesian A/B-test Calculator - What is the probability that your test variation beats the original? (<https://abtestguide.com/bayesian/>)

False Discovery Rate Calculator (<https://abtestguide.com/fdr/>)

Recommended reading

AB-testing tech note determining sample-size (<http://37signals.com/svn/posts/3004-ab-testing-tech-note-determining-sample-size>)

A clear picture of power and significance in AB-tests/ (<http://www.win-vector.com/blog/2014/05/a-clear-picture-of-power-and-significance-in-ab-tests/>)

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